

Rony Barua

Backend-Focused Full Stack Engineer

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SUMMARY

Full Stack Engineer with 4+ years of experience building scalable web applications using Node.js, React, and TypeScript. Experienced in API development, database design, authentication, real-time systems, testing, and CI/CD. Passionate about clean architecture, performance, and developer experience, with hands-on experience integrating AI solutions such as RAG and MCP.

SKILLS

Languages & Paradigms: JavaScript (ES6+), TypeScript, Python (basic), OOP, Functional Programming
Frontend: React JS, React Native, Next.js, Redux, Tailwind, Storybook, tRPC
Backend & Database: Node.js, Express.js, NestJS, GraphQL, REST API, Shopify API, MongoDB, PostgreSQL
Auth & Security: JWT, sessions & cookies, OAuth/OIDC, RBAC, secure secrets handling
Cloud & Tooling: Docker, AWS Lightsail, Redis, Vercel, Prisma, Stripe, Strapi, Socket.IO, Sentry, GitHub Actions
Testing & Quality: Jest, Vitest, React Testing Library, unit & integration testing, code review
AI & Automation: AI Integration, Workflow Automation (n8n), Basic RAG, MCP Server Integration

EXPERIENCE

Full Stack Engineer	Dec 2021 - Present
Corexlab Limited	Mirpur 10, Dhaka, Bangladesh
- Led 4+ full-stack projects, including the USEND Shopify dropshipping app, which is successfully running in production with 99.9% uptime and high scalability. - Contributing to Smaply (a journey mapping software), delivering multiple feature improvements across frontend and backend modules, and improving system usability. - Optimized complex database queries and implemented Redis-based caching, reducing API response times by up to 40% and improving performance, with CI/CD automation using GitHub Actions. - Wrote unit and integration tests (Jest, Vitest, React Testing Library) and performed peer code reviews to ensure correctness, security, and maintainability.	
Frontend Developer	July 2021 - Dec 2021
Flyte Solutions	Gulshan, Dhaka, Bangladesh
- Built responsive, pixel-perfect Next.js UIs for client projects, reducing load times by 20%. - Integrated RESTful APIs and optimized data fetching to cut latency by 20%. - Collaborated with designers and backend teams to deliver multiple production-ready features with improved UI responsiveness and user interaction flow.	

FEATURED PROJECTS

Natural Language Database Agent - Text-to-SQL AI Agent Live Demo
- Built a self-correcting AI agent that converts natural language to schema-aware SQL, executes it on Postgres, and returns explanations, tables, and charts with up to 3 error-aware retries. - Secured public access with SQL validation, read-only transactions, timeouts, row limits, and a SELECT-only role with automated privilege audits.

- Created a 43-question golden dataset evaluated by result-set accuracy, with GitHub Actions blocking regressions on pull requests.

Technologies: Next.js, TypeScript, PostgreSQL, Gemini API, Recharts

AI Knowledge Hub - RAG-Powered Knowledge Platform | **Live Demo**

- Engineered a full-stack Retrieval-Augmented Generation (**RAG**) platform enabling organizations to securely query internal knowledge with workspace-scoped RBAC and authentication.
- Architected a scalable document ingestion and indexing pipeline using vector embeddings and semantic search to deliver grounded AI responses with source citations and live indexing status.
- Implemented streaming AI chat and workspace-aware retrieval, providing accurate, traceable answers from organizational knowledge.

Technologies: Next.js, TypeScript, NestJS, PostgreSQL, Prisma, Qdrant, Redis, Gemini API, Tailwind CSS, tRPC

StyleVibe - Full Stack Salon Service App | **Live Demo**

- Built a full-stack salon booking platform with admin, seller, and customer portals in TypeScript, with **role-based access control (RBAC)** and JWT / session-cookie authentication.
- Integrated **Stripe Connect** to architect and manage secure, split transactions, including automated application fee deduction and instantaneous seller payouts.
- Implemented an **event-driven architecture** with queue-based booking and notification systems for handling transactions and disputes, enabling real-time updates and notifications. Integrated **Socket.IO** for real-time chat functionality.

Technologies: Next JS, Redux, RTK Query, Tailwind CSS, Express JS, MongoDB, Redis, Stripe, Socket.IO

CERTIFICATIONS

- Getting Started with **Docker** – Simplilearn (<https://bit.ly/4dlMxhS>)
- **Claude Code** in Action – Anthropic (<http://bit.ly/4t7oibT>)
- JavaScript Data Structures and Algorithms – freecodecamp.org (<https://bit.ly/48BWrcu>)

EDUCATION

B.Sc. in Computer Science & Engineering

Port City International University, Chittagong

CGPA- **3.72** / 4.00

Oct 2018